

From comparison to defensible temperature and salinity adjustments

A reusable scientific and software-development framework

Recommended scope. Build a systematic, profile-by-profile comparison of unadjusted MEOP temperature and salinity with independent reference profiles. Make the comparison reproducible, quality-controlled, uncertainty-aware, and usable at the scale of the MEOP archive. Once that foundation is validated, use it to produce *candidate* adjustment coefficients for expert review. Automatic modification of the operational coefficient tables is explicitly outside the scope of this framework.

Primary objective	A tested space–time matchup and comparison workflow, with documented outputs and a pilot scientific assessment
Optional extension	Conservative T/S adjustment suggestions with uncertainty and explicit accept/review/reject reasons
Existing foundation	Processed MEOP profiles, the CORA 202511 CTD/Argo reference archive, manifest-based acquisition and preparation, explicit batch statuses, coefficient application, and plotting/-batch machinery
Main scientific principle	Separate sensor bias from natural ocean variability, and never use an adjustment merely because a regression returns a number
Snapshot date	August 22, 2026

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1 The decision and the project goal

1.1 Recommendation

The two possible work packages are not truly separate. A method that estimates temperature or salinity adjustments first needs a scientifically defensible comparison: which external profiles are relevant, which observations pass quality control, how the data are brought to a common vertical coordinate, and how much of a difference can be explained by normal spatial and temporal ocean variability.

The recommended sequence is therefore:

Primary goal: develop and validate a systematic comparison framework.

Gated extension: use the same framework to estimate candidate adjustments, but only after the comparison passes agreed scientific tests.

This scope has a useful minimum result even if adjustment estimation proves too ambitious: the comparison archive, coverage report, diagnostics and tested matchup code will remain directly useful for future delayed-mode work. Conversely, starting with a coefficient estimator without fixing the comparison would risk fitting ocean variability, reference-data artefacts, or corrections that have already been applied.

	Systematic comparison	Adjustment estimation
Scientific risk	Moderate; assumptions can be exposed and tested	Higher; natural variability can look like instrument drift
Standalone value	High even if adjustment estimation is deferred	High only if validation is convincing
Existing support	CORA loading, plots and batch runner	Coefficient format and coefficient application
Main missing element	Profile-level space-time matchups	Validated estimator, uncertainty and decision rules
Recommended role	Core work package	Gated extension after scientific review

1.2 Scientific questions

The project should answer the following questions in order:

1. For each usable MEOP profile, are there independent reference profiles close enough in space and time to make a meaningful comparison?
2. After applying quality control and interpolating to a common pressure grid, what are the robust MEOP-minus-reference temperature and salinity anomalies?
3. How do these anomalies vary with pressure, time, region, season, matching radius and reference selection?
4. Are the observed patterns compatible with a constant offset, a pressure-dependent offset, a time-varying salinity offset, or simply unresolved ocean variability?
5. When a correction is proposed, does it improve independent held-out comparisons and remain stable under reasonable changes to the matching parameters?

1.3 Definition of success

The work is successful when it delivers all of the following core items:

- a documented and tested CORA inventory/QC reader that preserves profile identity, location, time, quality flags and provenance;
- an anti-meridian-safe, configurable space–time matchup method;
- a stable comparison data product for each tag, containing raw and adjusted anomalies, coverage and reference-spread estimates;
- reproducible per-tag and multi-tag summaries and plots;
- synthetic tests plus a scientifically diverse pilot evaluation;
- a batch command that separates usable results, missing or insufficient reference data, invalid targets and genuine software failures;
- concise user and developer documentation, including known limitations.

Candidate adjustment coefficients are a useful extension, but not a condition for a successful comparison framework. They should only be produced after an explicit scientific review accepts the comparison results and uncertainty analysis.

2 Review of what is already in place

2.1 Operational processing foundation

The renovated toolbox is a Python package under `src/meop_process/`. It already owns the main MEOP processing path: raw ODV import, location processing, quality control, low- and high-resolution products, thermal-lag processing, adjustment application, diagnostics, metadata summaries, publishing helpers and resumable batch execution. The package exposes installed commands including `meop-process`, `meop-compare`, `meop-compare-batch` and `meop-publish`.

The repository test suite contains 223 collected tests in this snapshot. Relevant regressions cover adjustment application, CORA tile loading, calibration-plot generation, batch resumption and NetCDF comparison. This is a strong software base: new work should extend it rather than create a disconnected notebook-only method.

2.2 Local data inventory at the review date

The working data tree is already large enough for development and retrospective validation. Counts below describe the local checkout on August 22, 2026; they are a snapshot rather than a property of the software.

Item	Count
Deployment directories under <code>data/data_prof/</code>	310
Unique tags with processed profile files	2,717
<code>lr0</code> , <code>lr1</code> , <code>hr0</code> , <code>hr1</code> , or <code>hr2</code> files	2,717 of each
<code>fr0</code> and <code>fr1</code> files	182 of each
Rows in <code>table_coeff.csv</code>	2,789
Tags represented in <code>table_salinity_offsets.csv</code>	77
CORA 202511 populated tile files	540
CORA 202511 profiles	4,328,083: 2,605,655 Argo and 1,722,428 CTD
CORA 202511 coverage	CTD: 1953-04-09 to 2025-06-30; Argo: 1995-09-07 to 2025-06-30
Collected repository tests	223

2.3 Existing pieces directly relevant to the work

Component	Status	What it does today
Processed profile products	✓	One NetCDF file per tag and processing level under <code>data/data_prof/<deployment>/</code> .
Runtime reference configuration	✓	<code>configs.json</code> points <code>references.cora_dir</code> to <code>CORA_ctd_argo_202511_tiles</code> .
CORA tile reader	●	<code>reference/cora.py</code> inventories actual files, accepts padded and unpadded longitude names, and returns latitude, longitude, JULD, TEMP, PSAL and a shared pressure grid. Calibration loads the union of buffered cells along accepted positions, handles anti-meridian windows circularly, and indexes selected profile rows before materialising TEMP/PSAL. The reader does not yet expose the retained profile provenance required for auditable matchups.
CORA source acquisition	✓	<code>meop-cora-download</code> produced a complete, release-pinned CORA 202511 source archive with exact hashed CTD and Argo file lists and a download manifest.
CORA tile preparation	✓	<code>meop-cora-prepare</code> applied per-profile type/QC/value rules, normalised time and longitude, interpolated to the 10-dbar grid, retained source/type/branch metadata, and wrote the installed 540-tile archive and preparation manifest.
Calibration figures	●	<code>plotting/calibration.py</code> produces T/S context plots, salinity anomaly plots, pressure-band statistics and a linear pressure fit.
Single-tag command	✓	<code>meop-compare -plot1 <tag></code> selects a processed target file and writes calibration PNG/CSV outputs.
Batch command	✓	<code>meop-compare-batch</code> discovers tags, resumes version-compatible successes, runs in parallel, logs each tag, distinguishes data-availability outcomes from failures, and writes CSV/Markdown summaries.
Regression comparator	✓	<code>workflows/compare.py</code> and directory mode in <code>meop-compare</code> compare two NetCDF outputs for numerical/software regression. This is not an oceanographic matchup method.
Coefficient application	✓	<code>processing/adjustments.py</code> reads operator tables, applies T/S corrections and writes adjusted variables, errors and calibration metadata.
Coefficient estimation	●	Existing calibration CSVs contain preliminary suggested coefficients, but the scientific comparison used to obtain them is not yet a valid basis for operational decisions.
Time-varying salinity adjustment	✓	Existing table values are interpolated by profile index and applied. Their breakpoints are operator supplied; no estimator finds them.
Expert-review workflow	✗	No proposal table, confidence classification, before/after validation report or controlled approval/import step exists.

2.4 What the current calibration workflow actually computes

For a target tag, the existing command prefers `hr1`, then `lr1`, `hr2`, `hr0`, `lr0`, `fr1` and `fr0`. Target preflight removes non-finite/null-island positions and minor spatial components disconnected from a supported deployment component. It verifies target-only profile and depth support before opening CORA. The reader then loads only cells intersecting local 5-degree windows along accepted positions and indexes matching reference rows before materialising T/S arrays. It projects the retained CORA profiles to the target pressure grid and computes one CORA median at each pressure level. Every accepted target profile is compared with this same median profile.

The diagnostic CSV reports robust summaries in fixed pressure bands and an ordinary linear fit over 200–1000 dbar. If

$$\Delta T(p) = T_{\text{tag}}(p) - T_{\text{CORA,median}}(p) \simeq b_T + a_T p, \quad (1)$$

the CSV writes candidate values $T1 = 1000a_T$ and $T2 = b_T$, with analogous values for salinity. These are syntactically compatible with the coefficient table, but the reference median is not matched to each target profile in time or local space. The numerical compatibility must not be confused with scientific validation.

2.5 How adjustments are already represented and applied

The operational table `data/tables/table_coeff.csv` stores one row per tag with `T1`, `T2`, `S1`, `S2`, removal flags and a comment code. The Python implementation applies

$$T_{\text{adjusted}}(p) = T(p) - 10^{-3}T1p - T2, \quad (2)$$

$$S_{\text{adjusted}}(p, i) = S(p, i) - 10^{-3}S1p - S2 + \delta S(i), \quad (3)$$

where p is pressure in dbar, i is profile index, and $\delta S(i)$ is an optional time-varying series interpolated from up to four index/offset pairs in `table_salinity_offsets.csv`. The code also updates adjusted-error variables and NetCDF calibration metadata.

The local coefficient history is valuable validation material: 262 rows have non-zero `T2`, 285 have non-zero `S1`, 1,051 have non-zero `S2`, and 77 tags have time-varying salinity-offset records. These are expert decisions, not perfect ground truth. They can be used to form test strata and to investigate disagreement, but should not simply be learned and reproduced.

The deprecated MATLAB code retained under `legacy/toolbox_legacy/` likewise applies pre-existing coefficients and interpolated salinity offsets. It does not provide the missing statistically validated estimator. The old external calibration plotting call is referenced by the legacy wrapper, but its full implementation is not present in this repository.

2.6 CORA 202511 reference archive

The configured reference is the prepared CTD/Argo archive from Copernicus Marine product `INSITU_GLO_PHY_TS_DISCRETE_MY_013_001`, dataset version 202511. The download manifest records 34,556 source files: raw CORA CT/OC/TE candidates for CTD selection and EasyCORA PF files received from Argo DACs. The preparation manifest records 540 populated 10-degree tiles and 4,328,083 retained profiles: 2,605,655 Argo profiles with `PROBE_TYPE=5` and 1,722,428 CTD profiles with `PROBE_TYPE=2`.

CTD coverage is 1953-04-09 through 2025-06-30; Argo coverage is 1995-09-07 through 2025-06-30. Every tile uses one standard time origin and a 100-level pressure grid from 10 to 1000 dbar in

10-dbar increments. A complete structural audit found no retained profile with non-finite time or position, no profile outside its named tile, no unexpected probe type, no malformed pressure grid, no skipped source file, and no missing QC variable.

The preparation accepts level, position and time QC flags 1 and 2, prefers an adjusted whole-profile parameter when available, otherwise uses the raw parameter, and requires both usable temperature and salinity on the target grid. Of 5,271,991 profiles matching the requested platform types, 400,035 were excluded by the strict time/position policy and 543,873 lacked sufficient paired T/S support. These exclusions are recorded in the manifest and should remain part of the scientific provenance.

The acquisition and preparation workflow is reproducible:

- `meop-cora-download plan|download` first saves the complete remote inventory, then selects exact global file lists, records size, version and hashes, and downloads source files without mixing them with generated tiles;
- `meop-cora-prepare` writes MEOP-compatible 10-degree tiles and a generation manifest while retaining `REFERENCE_KIND`, `PROBE_TYPE`, platform/reference identifiers, source file/class/dataset and the selected raw/adjusted parameter branch.

EasyCORA provides best-estimate, vertically subsampled Argo values, which is suitable for the 10-dbar comparison tiles but must remain visible in provenance. Full commands, policies and manifest-review checks are in `docs/cora_download_and_preparation.md`.

2.7 Calibration batch status after the reference update

The calibration products must be regenerated after a reference archive changes. The current batch command discovers 2,665 tags in enabled deployments, or all 2,717 locally processed tags with `-include-disabled`. It writes per-tag context plots, an offset diagnostic PNG and a diagnostic CSV containing pressure-band summaries and preliminary T1/T2/S1/S2 estimates. Batch method version 4 distinguishes `no_reference`, `insufficient_target`, `insufficient_reference`, `invalid_target` and `failed` from successful results and refuses to reuse incompatible success state. Success requires, for both temperature and salinity, at least 10 contributing target profiles, at least three valid levels in the 400–600 dbar band, finite band and linear estimates, and actual comparison coverage spanning at least 200–600 dbar. A clean rebuild must remove calibration plots, offset CSVs and calibration batch state before running `meop-compare-batch -include-disabled -force`.

3 What still needs to be developed

3.1 The central missing capability

The toolbox needs to move from a *track-wide visual background* to an auditable *profile-level comparison*. Each target profile must have its own reference neighbourhood, with explicit spatial distance, time separation, reference identity, quality-control decisions and uncertainty. All downstream summaries and coefficient suggestions must be traceable to those matchups.

3.2 Specific gaps found during the review

Gap	Why it matters	Direction
No per-profile time/space matching	A broad regional median mixes seasons, fronts and water masses	Match every target profile separately with configurable distance/time limits

Gap	Why it matters	Direction
Adjusted variables are preferred	Existing coefficients can leak into an estimator intended to discover those coefficients	Compare explicit raw TEMP/PSAL ; carry adjusted values only as a separate evaluation branch
Prepared CORA provenance is not exposed by the calibration loader	Source class, platform identity and raw/adjusted branch cannot yet be audited downstream	Return retained identifiers and branch metadata with the numerical arrays
MEOP value and position/time QC are not applied in calibration	Fill-value removal alone is not sufficient	Centralise flag decoding and masks for target and reference data
Reference identity is discarded	Self-matches, duplicated profiles and source dependence cannot be audited	Retain platform, cycle, data centre/reference, mode and source file/profile index
Reference archive identity is not part of batch invalidation	A successful state can be reused after the reference data change	Include source/preparation manifest hashes in the method configuration and batch state
Reference time is not used by the calibration method	A track-wide median mixes observations from different seasons and years	Decode the normalised JULD coordinate, match in time and test known dates
One value per pressure level is treated as independent	Confidence becomes far too optimistic because levels within a profile are correlated	Bootstrap or split by profile/time block, not individual depth level
No held-out validation	A fit can improve the same data used to create it and still be wrong	Reserve reference profiles or time blocks for evaluation
No proposal/review boundary	A numerical output could accidentally become an operational correction	Write a separate immutable proposal artefact; never edit official tables automatically

3.3 Out of scope

The following are outside the scope of this framework:

- fully automatic delayed-mode approval of corrections;
- changing or reproducing the upstream CORA reanalysis/validation system;
- redesigning the full MEOP processing and publication chain;
- treating expert historical coefficients as unquestionable truth;
- a sophisticated front-following ocean state estimate for every region;
- correction of pressure, oxygen, chlorophyll or light sensors.

4 Existing and proposed data structures

4.1 MEOP target profiles

Processed files follow

```
data/data_prof/<deployment>/<smru_name>_<qf>_prof.nc
```


For example, an `hr1` file has dimensions `N_PROF` and `N_LEVELS`, with one-dimensional profile metadata and two-dimensional hydrographic variables:

Group	Typical variables	Use in this project
Profile identity	<code>CYCLE_NUMBER</code> , platform metadata	Stable target-profile key
Time and position	<code>JULD</code> , <code>LATITUDE</code> , <code>LONGITUDE</code> plus QC	Match centre and filters
Raw hydrography	<code>PRES</code> , <code>TEMP</code> , <code>PSAL</code> plus QC	Primary data for bias estimation
Adjusted hydrography	<code>PRES_ADJUSTED</code> , <code>TEMP_ADJUSTED</code> , <code>PSAL_ADJUSTED</code> plus QC/error	Before/after evaluation only
Calibration metadata	Calibration-coefficient string array	Provenance and consistency checks

Use `hr1` when available because thermal-lag and high-resolution processing are important upstream steps, but read the unadjusted `TEMP` and `PSAL` fields for coefficient estimation. The processing level, choice of raw/adjusted field and all QC rules must be stored in the output provenance.

4.2 CORA reference profiles

The CORA 202511 reference interface uses 540 populated 10-degree NetCDF tiles. The preparation command uses one time origin and preserves the following fields for later matchup work:

- profile identifiers: `PLATFORM_NUMBER`, `CYCLE_NUMBER` and `DC_REFERENCE`;
- source identifiers: `SOURCE_FILE`, `SOURCE_FILE_CLASS`, `SOURCE_DATASET`, `REFERENCE_KIND` and `PROBE_TYPE`;
- normalised `JULD`, `LATITUDE` and `LONGITUDE`;
- `DATA_MODE`, `WMO_INST_TYPE`, and explicit `TEMP_SOURCE`/`PSAL_SOURCE`/`PRES_SOURCE` branch metadata;
- QC-masked, interpolated `TEMP` and `PSAL`;
- the shared `PRES_GRID` coordinate.

An extended reader should return an `xarray.Dataset` or a small typed object that keeps these fields together and decodes each tile's time before concatenation. A plain dictionary of six arrays is no longer sufficient once matchups must be audited. Results must record the CORA source download and preparation manifest hashes.

4.3 Recommended comparison products

Avoid making one enormous table with a row for every profile, reference neighbour and pressure level. A two-part structure is easier to understand and more compact.

4.3.1 Pair index: one row per target/reference pair

Write a CSV for each tag or batch partition containing:

```
target_tag, target_cycle, target_time, target_lat, target_lon,
reference_source, reference_platform, reference_cycle, reference_time,
reference_lat, reference_lon, distance_km, time_days,
space_weight, time_weight, combined_weight, accepted, rejection_reason
```

This table answers “which observations influenced this comparison?” without duplicating vertical profiles. CSV is deliberately suggested because it is already supported and can be inspected without a specialised stack. If scale becomes a problem, Parquet can be considered later with an explicit dependency decision.

4.3.2 Profile comparison NetCDF: one composite per target profile

Write one NetCDF per tag with dimensions N_PROF and N_LEVELS. Suggested variables are:

Variable/group	Meaning
PRES_GRID	Common comparison pressure grid
TARGET_TEMP_RAW, TARGET_Psal_RAW	QC-masked target values used for estimation
TARGET_TEMP_ADJUSTED, TARGET_Psal_ADJUSTED	Optional operational values used only for evaluation
REFERENCE_TEMP, REFERENCE_Psal	Weighted robust reference composite for this target profile
REFERENCE_*_SPREAD	Robust spread among reference neighbours, for example weighted MAD
ANOMALY_*_RAW	Raw target minus reference
ANOMALY_*_ADJUSTED	Adjusted target minus reference
N_REFERENCE	Number of accepted reference profiles for each target
MEDIAN_DISTANCE_KM, MEDIAN_TIME_DAYS	Matchup quality summaries
MATCHUP_QC	Good, marginal, no reference, bad target metadata, or insufficient overlap

Global attributes should include software version/commit, creation time, source file, reference archive, variable policy, QC policy, pressure grid, matching thresholds, weighting formula and a serialised configuration hash.

4.3.3 Tag summary and coefficient proposal

A batch-level summary CSV should contain coverage and robust anomalies by tag, variable, pressure band and time period. A separate proposal CSV may contain:

```
smru_platform_code, variable, model, T1, T2, S1, S2,
ci_low, ci_high, n_target_profiles, n_reference_profiles,
coverage_fraction, validation_delta_mad, sensitivity_range,
decision, decision_reason, method_version, created_at
```

The decision field should be one of `candidate`, `review`, `insufficient_data` or `reject`. This file is evidence for an operator; it is not an alternative `table_coeff.csv`.

4.4 Suggested output location

Keep experimental products separate from standard plots until the method is approved:

```
data/comparisons/cora/<method_version>/
  by_tags/<deployment>/<tag>_pairs.csv
  by_tags/<deployment>/<tag>_comparison.nc
  by_tags/<deployment>/<tag>_summary.csv
runs/<timestamp>/summary.csv
runs/<timestamp>/summary.md
runs/<timestamp>/logs/<tag>.log
latest/status.json
```

```
proposals/coefficients_<timestamp>.csv
```

This mirrors the repository’s existing resumable batch conventions without mixing experimental coefficient suggestions into operational tables.

5 General scientific methodology

5.1 Step 1: define a representative pilot set

Begin with 6–10 tags, then expand to 30–50 before attempting the full archive. The pilot must deliberately include:

- zero-correction tags and tags with known constant corrections;
- at least one pressure-dependent coefficient;
- at least one time-varying salinity correction;
- contrasting regions, seasons, track lengths and profile counts;
- a missing-position or no-reference case to exercise failure reporting;
- one or more tags excluded or flagged by historical expert review.

Useful starting examples already present locally are shown below. They are chosen to exercise code paths, not because historical coefficients are exact truth.

Tag	Historical feature	Why useful
BM01-544-12	All fixed coefficients zero	Negative-control case; 183 profiles
ct88-225-12	S2=-0.4	Strong constant salinity correction; existing regression fixture; 306 profiles
ct62-01-10	Non-zero T and S pressure terms	Exercises recovery of small slopes; 324 profiles
ct84-533-12	Fixed coefficients plus time-varying salinity	Exercises separation of a tag-wide model from profile-index change; 321 profiles
ct153-Pendragon-19	Non-zero T2 and S2	Long 1,093-profile track; already an LR-only adjustment regression case
ct104-EF970-13	Small constant T/S corrections	Long Southern Ocean track with 1,171 profiles

Confirm the final pilot list after producing a concise inventory of reference coverage and historical decisions.

5.2 Step 2: choose explicit input policies

The comparison must make three choices explicit:

1. **Target processing level.** Start with `hr1` where present, with a documented fallback. Do not silently mix levels in one scientific result.
2. **Target variable branch.** Use raw `TEMP/PSAL` for estimating calibration bias. Carry adjusted variables separately to measure whether existing corrections improve external agreement.
3. **Reference variable branch.** Use the prepared CORA best-parameter policy: adjusted whole-profile values when available, otherwise raw. Preserve `TEMP_SOURCE`, `PSAL_SOURCE`,

PRES_SOURCE and DATA_MODE in every auditable result. A different branch policy requires a separately identified reference build and sensitivity comparison.

5.3 Step 3: quality control and independence

Decode byte, character and object QC arrays through one tested helper. At minimum:

- reject invalid target time or position and apply target T/S QC level by level;
- verify the reference manifest and enforce the prepared archive’s accepted QC policy, then reject any remaining non-finite reference values;
- reject missing/fill values and non-physical pressure ordering;
- require a minimum common pressure span and number of valid comparison levels;
- retain the reason for every rejection;
- exclude exact duplicates and verify that no MEOP observation is compared with itself through the reference archive.

The local `list_wmo_numbers.csv` and CORA fields such as PLATFORM_NUMBER and DC_REFERENCE can help with independence checks. If complete self-source exclusion cannot be guaranteed, that limitation must be visible in the result and discussed with the scientific reviewer before coefficient estimation.

5.4 Step 4: create space–time neighbours

Convert longitudes to a consistent convention and positions to unit-sphere Cartesian coordinates. A `scipy.spatial.cKDTree` search on the sphere, followed by an exact haversine distance, is efficient and naturally handles the anti-meridian. Time should be normalised to UTC before matching.

Reasonable *initial values for sensitivity testing*, not final scientific constants, are:

- maximum distance: 150 km;
- maximum absolute time difference: 30 days;
- maximum neighbours retained per target: 20;
- minimum accepted reference profiles per target: 3.

Repeat key results for 50/100/200 km and 10/30/60 days. A coefficient that changes substantially across these defensible settings should be marked for review, not reported as a confident adjustment.

5.5 Step 5: vertical interpolation and the reference composite

Use a documented common pressure grid and never extrapolate outside the observed range of a profile. Perform interpolation only after masking invalid values. Preserve a count and robust spread at every target profile and pressure level.

A simple starting weight for accepted reference neighbour j is

$$w_j = \exp \left[-\frac{1}{2} \left(\frac{d_j}{L} \right)^2 - \frac{1}{2} \left(\frac{\Delta t_j}{\tau} \right)^2 \right], \quad (4)$$

where d_j is distance, Δt_j is time separation, and L, τ are configurable scales. Compare a weighted mean with a weighted median or trimmed mean. The robust choice is preferable where enough profiles exist; the method should not pretend that a one-neighbour composite has well-determined reference uncertainty.

5.6 Step 6: compute and inspect anomalies

For each target profile i , pressure p , and variable X , compute

$$\Delta X_i(p) = X_{\text{MEOP},i}(p) - \hat{X}_{\text{ref},i}(p). \quad (5)$$

Summaries should retain profile structure. Useful diagnostics include:

- anomaly versus pressure, with median and interquartile/reference-spread bands;
- profile-median anomaly versus time/profile number;
- anomaly versus distance, time separation and number of neighbours;
- maps coloured by anomaly and matchup coverage;
- raw-versus-adjusted residual comparison;
- summaries by stable pressure bands and, where scientifically useful, by potential-density or water-mass class;
- results under each space/time sensitivity setting.

First explain visible structure as oceanography before labelling it sensor bias. Front crossings, seasonal cycles, shallow mixed layers, sea-ice regions and sparse reference coverage are all plausible causes of structured anomalies.

5.7 Step 7: optional robust coefficient estimation

Only after the comparison passes an explicit scientific review, fit the existing operational parameterisation to raw anomalies over agreed stable pressure ranges:

$$\Delta T_i(p) = 10^{-3} T1 p + T2 + \epsilon_{i,p}, \quad (6)$$

$$\Delta S_i(p) = 10^{-3} S1 p + S2 + \epsilon_{i,p}. \quad (7)$$

Use robust regression so a small number of water-mass mismatches cannot dominate the fit. `scipy.optimize.least_squares` with a robust loss, or a clearly implemented iteratively reweighted method, avoids adding a large machine-learning dependency.

Do not treat all depth samples as independent. Estimate confidence by resampling target profiles or multi-day blocks. Store the full coefficient distribution or at least its percentile interval. Use separate training and validation profiles or time blocks.

For time-varying salinity, first plot profile-level residuals after the fixed model. A piecewise constant/linear correction with very few change points may be investigated as a stretch task. Penalise extra breakpoints, require persistence, bootstrap breakpoint stability and respect the existing limit of four index/offset pairs if proposing an operationally compatible result.

5.8 Step 8: decision gates

A proposed coefficient should be labelled **candidate** only when all agreed gates pass. Starting gates to refine with the scientific review group are:

- enough matched target profiles and adequate deployment-time coverage;
- more than one reference platform and reasonable spatial/time separations;
- sufficient vertical overlap in the chosen pressure band;

- a bootstrap interval narrow enough for the intended correction;
- the sign and magnitude are stable across matching sensitivities;
- held-out median bias and MAD improve, not just training fit;
- no implausible T/S or density behaviour is introduced;
- results are understandable on the diagnostic plots.

Failure of a gate is useful information. The correct output may be `insufficient_data` or `review`; the estimator must never force every tag to have a non-zero correction.

6 Validation plan

6.1 Synthetic recovery tests

Create small synthetic MEOP/CORA fixtures with known offsets, pressure slopes, missing levels, bad QC, time offsets, longitude wrap and a known change point. Tests should show that the pipeline:

- selects exactly the expected neighbours;
- handles 179 degrees east versus 179 degrees west correctly;
- never extrapolates vertically;
- recovers injected T1/T2/S1/S2 within a stated tolerance;
- does not infer a correction in the zero-bias case;
- reports no-reference and insufficient-data cases honestly;
- widens uncertainty when reference coverage is poor.

6.2 Historical comparison

Stratify historical tags into zero coefficient, constant offset, pressure-dependent, time-varying, salinity removed and fully removed groups. For each stratum, compare candidate results with the operator tables, while recognising that disagreement can mean the new method is wrong, the reference is unsuitable, or the historical decision used evidence not available to the algorithm.

Report at least coefficient sign agreement, robust difference, interval coverage, false-positive rate on zero-coefficient tags, and the fraction classified as insufficient. Review individual large disagreements rather than hiding them in one correlation coefficient.

6.3 Held-out and sensitivity validation

Fit on one part of the deployment and evaluate on another, or divide independent reference platforms between fit and evaluation. Compare before/after median anomaly and MAD. Repeat under several matching windows and reference-variable policies. The most important result is not the smallest residual; it is a correction that generalises and remains stable when defensible assumptions change.

6.4 Scientific review gate

Before coefficient estimation begins, review:

1. corrected CORA coverage and explicit batch status counts;
2. pair maps and anomaly plots for the pilot tags;

3. results of matching-window sensitivity tests;
4. evidence that raw and adjusted branches are not mixed;
5. a recommendation to proceed with fixed coefficient estimation, limit the project to comparison, or revise the reference strategy.

Proceeding to estimation should be a scientific decision made at this review, not an automatic consequence of completing the comparison software.

7 Final perspective

Most of the engineering substrate is already present: processed MEOP profiles, coefficient application, CORA tiles, plotting, batch state and tests. The missing piece is the scientific middle layer that turns “profiles in the same broad box” into traceable, independent and uncertainty-aware comparisons.

That layer is a coherent scientific development project spanning data handling, geospatial algorithms, oceanographic interpretation, statistics, testing and reproducible software development. The systematic comparison is useful in its own right and provides the evidence needed to decide how far defensible adjustment estimation can go.

References

- [1] Roquet, F., Charrassin, J.-B., Marchand, S., Boehme, L., Fedak, M., Reverdin, G., and Guinet, C. (2011). Validating hydrographic data obtained from seal-borne satellite-relayed data loggers. *Journal of Atmospheric and Oceanic Technology*, 28, 787–801. doi:10.1175/2010JTECHO801.1.
- [2] MEOP/Sea Mammal data-processing manual, version 1.2, available locally as docs/seamammal_user_manual_version1.2.pdf.
- [3] MEOP Process renovation status and toolbox guide, available locally as docs/meop_toolbox_status.tex.
- [4] Copernicus Marine Service, Product User Manual for INSITU_GLO_PHY_TS_DISCRETE_MY_013_001, issue 1.17 (November 2025). [Official CORA user manual](#).
- [5] Copernicus Marine Service, Global Ocean CORA in-situ observations, delayed mode. [Product and data access page](#).